



Analysis and evaluation of antibacterial effects of new herbal formulas, AP-001 and AP-002, against *Escherichia coli* O157:H7

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ABSTRACT

Aims: Over the last few years, there has been an increasing interest in the use of natural substances for promoting human and animal health. Antibiotic resistance, enhanced by excessive use of antibiotics, has encouraged human and animal health authorities to consider and employ alternative approaches, including herbal and botanical medicine to combat invading microorganisms. Herein, the essential oils of *Thymus vulgaris*, *Allium cepa*, *Allium sativum*, *Eucalyptus globulus*, *Salvia officinalis*, *Dianthus caryophyllus*, *Mentha spicata* and *Mentha piperita*, were evaluated for their antibacterial activities against standard *Escherichia coli* O157:H7 by disk diffusion method.

Main methods: Minimum Inhibitory Concentrations (MICs) and Minimum Bactericidal Concentrations (MBCs) were assessed via microdilution assay. In the next step, two combinations of these essential oils were formulated. Their in-vitro antibacterial effects were evaluated and compared with a commercially available herbal drug containing *T. vulgaris* essential oil.

Key findings: According to the results, the two new formulated essential oil combinations had more potent antibacterial effects against *E. coli* when compared to the commercial herbal drug.

Significance: The presented data indicate the potential antibacterial activity of these newly formulated essential oil remedies to be employed in the poultry industry in the fight against colibacillosis, although this claim has to be examined in experimental and clinical trials.

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1. Introduction

Traditional medicine has been used for thousands of years. Nowadays in some Asian and African countries, more than 80% of the populations rely on this type of medicine for primary health care. Alternative medicine using herbal and botanical substances can be a lucrative business with the capacity to generate billions of dollars in revenue. Currently, more than 100 countries have regulations for herbal medicines [39].

Plants produce a high diversity of secondary metabolites with important function of protecting them against many biological threats such as microbial pathogens. Almost 100,000 secondary metabolites are known and it is believed that most of them are involved in plant chemical defense system [6]. More than 3000 essential oils are known, of which about 300 are commercially important and used by the flavor and fragrance industries [34].

Essential oils (EOs) are aromatic oily liquids derived from plant material (flowers, buds, seeds, leaves, twigs, bark, herbs, wood, fruits and

roots). They have several properties such as antibacterial [17,29,37], antiviral [3,12,40], antimycotic [2,17], antitoxigenic, antiparasitic [26,31] and insecticidal [15] properties.

In recent years, widespread bacterial resistance to most synthetic antibiotics (especially in the poultry industry) along with its impact on the environment has led health authorities to use herbal and botanical medicine in combating bacterial infections in animal industry.

A shrewd approach employed to overcome antimicrobial resistance mechanisms is the use of combinations of drugs and several plant extracts with synergistic activity against pathogenic microorganisms [24]. Antimicrobial activity of essential oils or so called volatile oils has been confirmed in many studies. In-vitro antimicrobial activities of essential oils from various herbal families such as the Labiatae, Lamiaceae and Alliaceae have been reported [8,14,28]. Also, several studies reported their essential oil antibacterial activities against pathogenic bacteria such as *Escherichia coli* [7,36].

The present study aimed to investigate the antimicrobial effects of 8 essential oils (obtained from *Thymus vulgaris*, *Allium cepa*, *Allium sativum*, *Eucalyptus globulus*, *Salvia officinalis*, *Dianthus caryophyllus*, *Mentha spicata* and *Mentha piperita*) and their selective combinations as two separate formulations (AP-001 and AP-002) against standard *E. coli* O157:H7.

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2. Materials and methods

2.1. Plant material and extraction of essential oils

Based on the literature, eight herbs with potential antibacterial effects, including *A. cepa* (bulb onion or common onion), *D. caryophyllus* (carnation or clove pink), *T. vulgaris* (thyme, common thyme, garden thyme), *M. spicata* (spearmint or spear mint), *S. officinalis* (sage, garden sage or common sage), *M. piperita* (peppermint), *A. sativum* (garlic) and *E. globulus* (Tasmanian Blue Gum, Southern Blue Gum or Blue Gum) were selected (all donated by Johare Taem Co., Mashhad, Iran; and properly identified by the Ferdowsi University of Mashhad Herbarium (FUMH) in June 2011) and grounded up into a fine powder and put in a 1000 ml distillation flask. A 500 ml aliquot of distilled water was added and volatile oil distillation apparatus was set according to Chinese Pharmacopoeia [11]. The resultant mixture was distilled for 6 h, while the distillate was collected from the condenser and dried over anhydrous sodium sulfate. The obtained essential oil was stored at room temperature until further analysis.

2.2. Microbial strain and preparation

The bacterial strain (*E. coli* O157:H7; ATCC 43895) was used for antibacterial assessment of various essential oils and their selective combinations throughout the experiments. The bacterial suspensions used in all experiments were adjusted to a turbidity equivalent to a 0.5 McFarland standard.

2.3. Disk preparation

Paper blank disks (Pad Tan Teb Co.) were impregnated with each essential oil or the formulated combinations of essential oils for 15 min and dried at 37 °C for 20 min. Negative control disks underwent the same process in the tube containing distilled sterile water, while gentamycin impregnated disk was used as positive control.

2.4. Disk diffusion method

Antibacterial activities of the 8 essential oils, as well as the EO combinations (AP-001, AP-002 and Antibiofin®) were assessed using the paper disk agar diffusion method as reported by Sacchetti et al. [35]. The bacterial suspensions were inoculated on the surface of Mueller Hinton agar plates. Subsequently, the EO containing disks and related negative (distilled sterile water) and positive (gentamycin) control disks were placed on the cultures. Plates were then incubated overnight at 37 °C for 24 h. After the incubation period, the inhibition zone diameters were measured in millimeters and compared to those of the positive controls. All the tests were performed in duplicate.

2.5. Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC) (broth microdilution method)

MICs and MBCs of the EOs and the formulated combinations (AP-001, AP-002), as well as Antibiofin were determined by microdilution assay using liquid cultures in 96-well micro plates [9]. Using Mueller Hinton broth, a series of twofold dilutions of each sample were prepared in sterile 96-well micro plates. Fifty microliters of each dilution, ranging from 1000 to 1.9 (μl/ml), was mixed with an equal volume of bacterial suspension and Mueller Hinton broth. Bacterial (100 μl broth and 50 μl bacteria suspension) and blank (150 μl broth) growth controls were prepared, while gentamycin was considered as the antibacterial positive control. Plates were incubated at 37 °C for 24 h under normal atmospheric conditions. The MIC was defined as the lowest concentration (highest dilution) of the herbal essential oils that inhibited the visible growth (no bacterial growth sedimentation or less than positive control). Afterwards, 10 μl of each well was transferred to Mueller Hinton

agar plates and incubated at 37 °C for 24 h. The lowest concentration associated with no visible growth of bacteria on the agar plates was considered as MBC. All dilutions were performed in duplicate.

2.6. Gas Chromatography–Mass Spectrometry (GC–MS) analysis

Analysis and desorption of volatile compounds of four EOs that showed the greatest antibacterial properties (*T. vulgaris*, *M. piperita*, *D. caryophyllus* and *E. globulus*) were performed using a Hewlett Packard 6890 GC system, coupled with a Finnigan MAT Incos-50 instrument mass selective detector. Compounds were analyzed within the validated stability timeframe, separated on a DB-5 fused silica capillary column (25 m × 0.25 mm, film thickness 0.25 μm) and identified by the comparison of their retention indices (RI) relative to n-alkanes with literature values. Split injection was conducted with a ratio of 25:1 for distillation sample, while splitless injection was used for the SPME sample. The column oven temperature was programmed to rise from an initial temperature of 60 °C (2 min) to 150 °C at 15 °C/min, then to 200 °C at 3 °C/min and finally to 280 °C at 4 °C/min, while this final temperature was maintained for 2 min. The injection temperature and ion source temperature were 280 °C and 150 °C, respectively. Helium with a flow rate of 1.5 ml/min was used as the carrier gas. The ionizing energy was 70 eV and the ionization current was 2 A. The data were acquired by collecting the full scan mass spectra within the scan range from 40 to 500 amu. Compounds were tentatively identified using the NIST Mass Spectral Search Program (National Institute of Standards and Technology, Washington, D.C., USA) [18].

2.7. Preparation of formulations

Among the eight herbals used, four essential oils were selected according to the qualitative and quantitative antibacterial properties obtained in the first phase of the study. Two formulations (AP-001 and AP-002) were designed according to the molecular structure and physico-chemical properties of the active constituents of these essential oils. Both formulations contained 90% (V/V) of the mixture of four plants (*E. globulus*, *D. caryophyllus*, *M. piperita* and *T. vulgaris*) essential oils, with different proportions. The remaining 10% (V/V) included excipients as emulsifiers (DMSO and propylene glycol) (Table 1). The antimicrobial properties of these formulations, MICs and MBCs, were assessed using disk diffusion and microdilution methods, while a commercially available formulation (Antibiofin®, Pars Imen Daroo, Iran, containing *T. vulgaris*-derived essential oil) was used for comparison.

2.8. Statistical analysis

The diameters of the inhibition zones in disk diffusion method, as well as values of MICs and MBCs assessed by microdilution assay were analyzed by one-way ANOVA followed by Tukey–Kramer multiple comparison test (GraphPad InStat Version 3.0, GraphPad Software Inc., USA). $p < 0.05$ was considered significant.

Table 1
Composition of formulated EO combinations and Antibiofin (%).

Component		AP-001	AP-002	Antibiofin
EOs	<i>Eucalyptus globulus</i>	50	27	–
	<i>Dianthus caryophyllus</i>	25	50	–
	<i>Menthe piperita</i>	11	9	–
	<i>Thymus vulgaris</i>	4	4	100
Excipients	DMSO and propylene glycol	10	10	–
Total		100	100	100

3. Results

3.1. Comparative growth inhibitory properties of essential oils

To assess the comparative susceptibility of the bacterial strain (*E. coli*), the growth inhibition zones measured by disk diffusion method were assessed for the 8 different herbal essential oils (Table 2). The results indicated that EOs of *E. globulus*, *T. vulgaris* and *D. caryophyllus* produced the most potent bacterial growth inhibitory properties, while *A. sativum*, *A. cepa*, and *S. officinalis* EOs showed no significant inhibition zones.

3.2. Antibacterial potency of essential oils

The MICs and MBCs were measured to assess the potency of EOs. Microdilution assay revealed that *D. caryophyllus*, *T. vulgaris*, *M. spicata*, *M. piperita* and *E. globulus* possess potent bacteriostatic and bactericidal effects (Table 3). Consistent with the results from disk diffusion assay, the high MICs and MBCs obtained by *A. cepa*, *S. officinalis* and *A. sativum* revealed weak potency of these EOs to inhibit the bacterial growth and viability.

3.3. Major phytochemicals in the formulations

Considering the preliminary results of bacteriostatic and bactericidal properties, four EOs were selected to be formulated as EO combinations. Major bioactive constituents of EOs extracted from *T. vulgaris*, *M. piperita*, *D. caryophyllus* and *E. globulus* were determined by GC/MS analysis. Among nineteen (19) classes of phytochemicals, 1,8-cineol (eucalyptol), thymol, eugenol, L-menthol, L-menthone, D-limonene and carvacrol were the major constituents (Table 4). These phytochemicals were found to be stable over 12 months at room temperatures (25 °C to 30 °C) indicating a shelf life of not less than 12 months.

3.4. Comparative growth inhibitory properties of formulations

The growth inhibition zones achieved by two formulations against *E. coli* cultures were assessed using disk diffusion method (Fig. 1). Comparative susceptibility test indicated that formulations AP-001 and AP-002 produced a wider zone of bacterial growth inhibition compared to the commercially available Antibiofin® formulation (Table 5).

3.5. Antibacterial potency of formulations

The antimicrobial activity of two formulations was compared with that of a commercial herbal antimicrobial agent, Antibiofin®. Using

Table 2
Antimicrobial activity of investigated essential oils against the tested microorganism.

Samples		Inhibition zone diameter ^a (mm)
		<i>E. coli</i> O157:H7
EOs	<i>Allium cepa</i>	nd ^b
	<i>Dianthus caryophyllus</i>	16.5 ± 0.5
	<i>Thymus vulgaris</i>	17.0 ± 2.8
	<i>Mentha spicata</i>	10.0 ± 3.4***
	<i>Salvia officinalis</i>	nd
	<i>Mentha piperita</i>	9.50 ± 2.0***
	<i>Allium sativum</i>	nd
	<i>Eucalyptus globulus</i>	21.0 ± 1.7
	GM ^c	21.7 ± 0.5
	SDW ^d	nd
Controls		

^a Results presented as mean ± standard deviation from the experiments in duplicate.

***, $P < 0.001$ vs. GM. The diameter of the disks ($\varnothing = 6$ mm) was included.

^b nd: not detected.

^c GM: gentamycin (10 µg/disk); positive control.

^d SDW: sterile distilled water; negative control.

Table 3

Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC) (µl/ml) of selected plant essential oils against the tested microorganism.

EOs	Strain tested	
	<i>E. coli</i> O157:H7	
	MIC ^a	MBC ^a
<i>Allium cepa</i>	93.8 ± 44.2	312.5 ± 265**
<i>Dianthus caryophyllus</i>	19.5 ± 16.5	46.9 ± 22.1
<i>Thymus vulgaris</i>	62.5 ± 0.00	93.8 ± 44.2
<i>Mentha spicata</i>	46.9 ± 22.1	78.1 ± 66.3
<i>Salvia officinalis</i>	187.5 ± 88.4*	375 ± 176**
<i>Mentha piperita</i>	46.9 ± 22.1	56.9 ± 22.1
<i>Allium sativum</i>	500 ± 0.00***	>500***
<i>Eucalyptus globulus</i>	23.4 ± 11	62.5 ± 0.00
GM ^b	1.9 ± 0.0	2.9 ± 1.4
SDW ^c	nd ^d	nd

* $P < 0.05$ vs. GM.

** $P < 0.01$ vs. GM.

*** $P < 0.001$ vs. GM.

^a Results are presented as mean ± standard deviation from the experiments in duplicate.

^b GM: gentamycin; positive control.

^c SDW: Sterile distilled water; negative control.

^d nd: not detected.

microdilution assay, the MIC and MBC values of these formulated combinations indicated the statistically significant higher bactericidal, as well as bacterial growth inhibitory activities against *E. coli* (Table 6).

4. Discussion

The growing extent of administering antimicrobial agents has created a great concern worldwide, intimidating not only public health, but also clinical therapeutic success rate via increased incidence of bacterial strains that are resistant to antimicrobial agents. Concerning drug residues in food, this might gain more importance when it comes to uncontrolled administration of these agents as feed additives, growth promoters or therapeutic antibacterial drugs in food producing animals. Nowadays, the immense administration of antimicrobial agents in both public health and veterinary sectors, especially in the poultry industry has caused a selection pressure that favors the survival of antibiotic resistant strains and has emerged as a global problem of multiple drug resistance (MDR).

Table 4

Essential oil composition of *Thymus vulgaris*, *Dianthus caryophyllus*, *Eucalyptus globulus* and *Mentha piperita* determined by GC–MS analysis.

Compound	<i>T. vulgaris</i> (%)	<i>M. piperita</i> (%)	<i>D. caryophyllus</i> (%)	<i>E. globulus</i> (%)
D-Limonene	–	–	–	8.5
Alpha-pinene	–	0.5	–	6
Beta-pinene	–	1	–	0.5
Alpha-Phellandrene	–	–	–	0.5
Eucalyptol (1,8-Cineol)	–	–	–	84.5
Eugenol	–	–	88	–
Eugenol acetate	–	–	9.5	–
Caryophyllene	–	–	3.14	–
L-Menthol	–	46.5	–	–
L-Menthone	–	30	–	–
L-Menthyl acetate	–	2.5	–	–
Cineol	–	7	–	–
Limonene	–	3	–	–
Thymol	25.28	–	–	–
Carvacrol	51.27	–	–	–
Linalol	18.51	–	–	–
Beta myrcene	2.46	–	–	–
Gamma terpinene	5.51	–	–	–
Para cymene	3.14	–	–	–

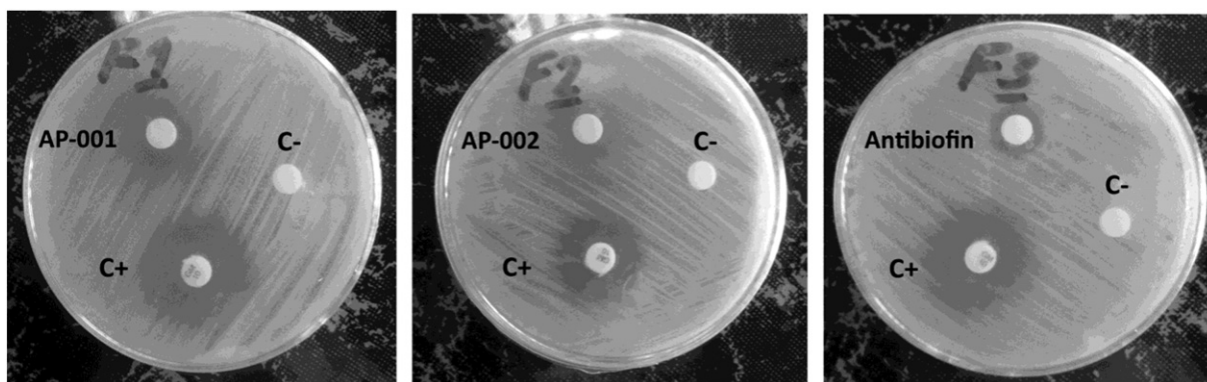


Fig. 1. Growth inhibition zone of *E. coli* induced by the two formulations (AP-001 & AP-002) and commercial compound (F3) in disk diffusion method; C– : negative control, distilled sterile water; C+ : positive control, gentamycin; the diameter of the disks ($\varnothing = 6$ mm) was included.

New trends in drug discovery emphasize on natural resources with antimicrobial properties. A huge body of studies reports the antimicrobial effects of different herbals, especially those with substantial volatile essential oil content, including thyme, peppermint, clove, mint, garlic, and eucalyptus [4,14,17,32].

Higher antimicrobial activity of thyme, eucalyptus, clove and peppermint-derived EOs in this study was evident in disk diffusion assay, consistent with the MIC and MBC results obtained from broth microdilution assay. It can be explained by the presence of significant amounts of carvacrol, eugenol, D-limonene, eucalyptol and thymol. It has been known that antimicrobial activity of essential oils (EOs), or aromatic plant essences, as volatile and fragrant substances with an oily consistency, is attributed to the presence of low molecular weight phenols, terpenes and aldoketones [7,10,13,25]. There are also reports elucidating that the antimicrobial activity of EOs is due to their content of oxygenated terpenoids (e.g., alcohols and phenolic terpenes), while some hydrocarbons also exhibit antimicrobial effects [7,13,25].

Herein, we have validated our results by measuring the most active components of EOs used in formulated combination forms. In line with our results, Ghasemi Pirbalouti et al. have identified twenty four components in *T. vulgaris*, indicating thymol, carvacrol, α -terpinene and p-cymene as major constituents [19]. Other reports have also emphasized that in particular, carvacrol and thymol are the major compounds of thyme essential oil. The significant in vitro antimicrobial activity of these two phenolic monoterpenes and isomers has been already studied and reported against several bacterial strains [33].

GC/MS analysis of *E. globulus* has also identified 1,8-cineole as the main compound with the highest antibacterial activity on the Gram positive *Staphylococcus aureus* [16]. It has also been shown that oxygenated monoterpenes, 1,8-cineole and α -terpineol possess relatively strong antimicrobial properties against many important pathogens [1].

The analytical results of the present study underlying menthol, menthone, cineol, limonene and menthyl acetate as the most abundant

components of *M. piperita* EO, were consistent with other reports. It has been shown that the antimicrobial activity of peppermint essential oil could be correlated to the presence of menthol and menthone. 47 components were identified in *M. piperita* oil, while the major components of the oil were menthol, iso-menthone, limonene, iso-menthanol and menthyl acetate [38].

Eugenol is the major component of the *D. caryophyllus* essential oil. It showed strong antibacterial effects against most tested pathogens, such as *Staphylococcus aureus*, *Proteus mirabilis* and *E. coli*, while the best MIC values were reported against *Bacillus cereus*, *Listeria monocytogenes* and *Klebsiella pneumonia* [30].

However, in practical terms, limited success has been recorded in using these herbal extracts as alternatives to chemical antibiotics. This may indicate the imperative needs for empowering these agents via different pharmacological approaches such as using their most active extract forms, or combining them to achieve latent synergistic effects. Furthermore, other minor components, in relatively low concentrations, such as α -pinene, β -pinene, and γ -terpinene have also a significant impact on antibacterial activity, possibly by producing a synergistic effect to the major components [7]. Although there are some concerns about the possibility of antagonism between different components, several studies have reported that whole EOs usually have higher antibacterial activity compared to their major components, probably due to potential additive or synergistic effects.

The additive effects of oregano and thyme have been reported against *Listeria* strains [22]. There are also promising results indicating enhanced antibacterial effects against *B. cereus*, *E. coli*, *L. monocytogenes* and *P. aeruginosa* achieved by oregano in combination with basil, thyme or marjoram [21]. Furthermore, synergism has

Table 5
Antimicrobial activity of the two formulas (AP-001, AP-002) against *E. coli*.

Samples		Inhibition zone diameter ^a (mm)
		<i>E. coli</i> O157:H7
Formulas	AP-001	13.5 $\pm$ 0.7
	AP-002	13.0 $\pm$ 0.0
	Antibiofin	12.0 $\pm$ 0.0
	GM ^b	19.5 $\pm$ 0.9
Controls	SDW ^c	nd ^d

^a Results are presented as mean $\pm$ standard deviation from the experiments in duplicate; The diameter of the disks ($\varnothing = 6$ mm) was included.

^b GM: gentamycin (10 μ g/disk); positive control.

^c SDW: sterile distilled water; negative control.

^d nd: not detected.

Table 6

Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC) (μ l/ml) of the two formulas (AP-001, AP-002) against *E. coli*.

Samples	Strain tested	
	<i>E. coli</i> O157:H7	
	MIC ^a	MBC ^a
AP-001	3.90 $\pm$ 0.0	7.8 $\pm$ 0.00
AP-002	2.90 $\pm$ 1.4	5.8 $\pm$ 2.7
Antibiofin	31.2 $\pm$ 0.0***	46.8 $\pm$ 22.1*
GM ^b	1.9 $\pm$ 0.0	2.9 $\pm$ 1.4
SDW ^c	nd ^d	nd

* $P < 0.05$ vs. GM.

*** $P < 0.001$ vs. GM.

^a Results are presented as mean $\pm$ standard deviation from the experiments in duplicate.

^b GM: gentamycin; positive control.

^c SDW: sterile distilled water; negative control.

^d nd: not detected.

been observed between carvacrol and its precursor p-cymene and between cinnamaldehyde and eugenol. In fact, carvacrol and thymol can work together to disintegrate the outer membrane of Gram-negative bacteria, releasing lipopolysaccharides (LPS) and increasing the permeability of the cytoplasmic membrane to ATP [7].

Therefore, a compelling strategy for formulating a potent antibacterial agent to control pathogenic microorganisms is the use of whole EOs instead of purified or artificial synthetic components. On the other hand, formulated combinations, either single EOs or mixtures of purified main components, produce a plethora of interactive antibacterial effects via affecting multiple biochemical processes in the bacteria [13,23]. In the present study, the improved antibacterial activity was obvious when formulated combinations of the selected EOs were compared to the commercially available EO of *T. vulgaris*. We should consider the fact that in qualitative disk diffusion method, minor differences in diameters of the bacterial growth inhibition zone might reflect huge variations in antibacterial activity profile, changing the drug antibacterial category between resistant, moderate and sensitive. We further validated our results by using the MIC and MBC quantitative antibacterial assays, indicating higher antibacterial potency of our newly formulated EO combinations compared to Antibiofin®.

5. Conclusion

Indeed, our data revealed a powerful antibacterial activity of two formulations containing selective EOs compared to commercially available formulation composed of a single herbal EO (derived from *T. vulgaris*). Although the findings are still preliminary, it appears to hold promise of a new herbal antibacterial with potential effects especially against poultry pathogens. Further assessment of these formulations is needed to be performed by in vivo experiment, including laboratory experiments and clinical field trials in poultry farms.

Conflict of interest statement

The authors declare that there are no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.lfs.2015.05.007>.

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